

# Double Pendulum and The Chaos Theory

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May-June, 2026

# Related CALCAB Knowledge

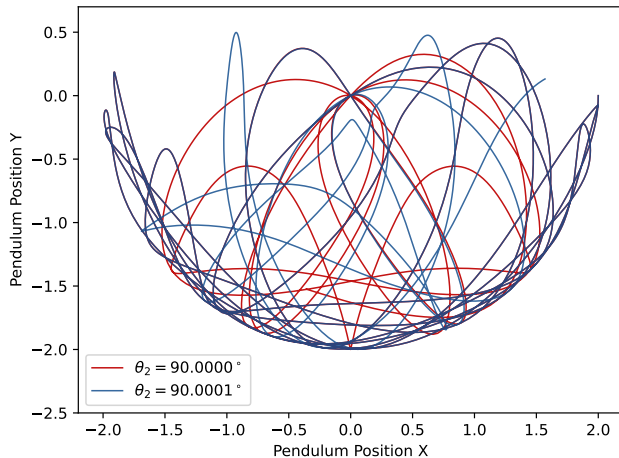
**Differential Equation:** It is the core of simulation algorithm we applied in this project, the entire simulation was run based on a differential equation derived by *Lagrangian Mechanics*, so the differential equation we learned in unit 7 is the core of our project.

**Limit and Approaching:** In our project, the observation of the simulation accuracy as time step changes is also a core part. By speculating the result approaches a value as step decreases (becomes more accurate), we could choose a suitable time step for simulation that is both efficient and accurate.

**Integration (by Part):** For deriving Euler-Lagrange Equation.

# Introduction To The Chaos Theory

Two Slightly Different Pendula Simulation Outcome



**Uncertain Outcome:** In a chaotic system described in the Chaos Theory, even a *negligible* disturbance or initial differences could cause an incredibly different result.

**Determinable though Chaotic:** Chaotic doesn't mean random and indeterminate, instead, chaotic systems could be predicted by calculations and modeling if one has incredibly precise data, though the result is input-sensitive.

# Double Pendulum, Where Newtonian Physics Dies

# Lagrangian Physics and Double Pendulum

Start from the Euler-Lagrangian Equation,

$$\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = 0 \quad (1)$$

where the Lagrangian  $L = T - V$ ,  $\partial$  (in multivariable calculus) basically means what  $d$  represents in normal calculus,  $q$  is the generalized coordinates, which is angle ( $\theta_i$ ) in our problem.

Citing 1, for double pendulums, we have

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\theta}_1} \right) - \frac{\partial L}{\partial \theta_1} = 0, \quad \frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\theta}_2} \right) - \frac{\partial L}{\partial \theta_2} = 0 \quad (2)$$

# Lagrangian Physics and Double Pendulum

Now let's move on to physics, deriving expressions for energy.

Position of Ball 1:  $x_1 = l_1 \sin \theta_1$ ,  $y_1 = -l_1 \cos \theta_1$

Position of Ball 2:  $x_2 = x_1 + l_2 \sin \theta_2$ ,  $y_2 = y_1 - l_2 \cos \theta_2$

So,

Speed of Ball 1  $v_1 = \frac{dx_1}{dt}$ , Speed of Ball 2  $v_2 = \frac{dx_2}{dt}$

Total  $E_k = \frac{1}{2}m_1(v_1)^2 + \frac{1}{2}m_2(v_2)^2$ , Total  $E_p = m_1gy_1 + m_2gy_2$

$$L = T - V = E_k - E_p = \frac{1}{2}m_1(v_1)^2 + \frac{1}{2}m_2(v_2)^2 - m_1gy_1 - m_2gy_2$$

# Introduction To DE Calculation Approaches

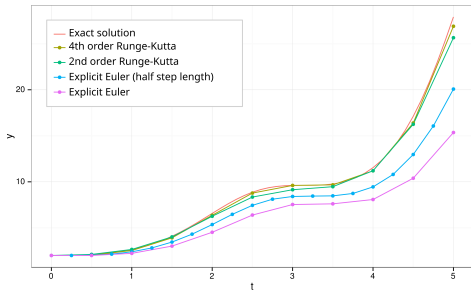


Figure: Comparison of the Runge-Kutta methods for the differential equation. Red line is the exact solution, 4th order Runge-Kutta is what we mainly use in this project, and is significantly more accurate. Figure from Wikimedia Commons, CC BY-SA 3.0 

## Fourth-Order Runge-Kutta (RK4) Approach:

$$\text{Let } \frac{dy}{dx} = f(x, y),$$

$$y_{n+1} = y_n + \frac{\Delta x}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Where,

$$k_1 = f(x_n, y_n)$$

$$k_2 = f\left(x_n + \frac{\Delta x}{2}, y_n + \frac{\Delta x}{2}k_1\right)$$

$$k_3 = f\left(x_n + \frac{\Delta x}{2}, y_n + \frac{\Delta x}{2}k_2\right)$$

$$k_4 = f(x_n + \Delta x, y_n + \Delta x k_3)$$

# Euler's Method: Simple DE Approximation

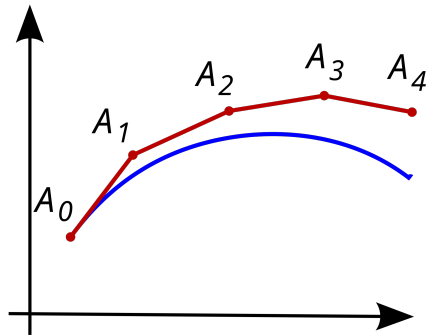


Figure: Illustration of the Euler method. The unknown curve is in blue, and its Euler approximation is in red. Figure from Wikimedia Commons, Public Domain. 

Euler's Method:

$$\text{Let } \frac{dy}{dx} = f(x, y),$$

$$y_{n+1} = y_n + \Delta x f(x_n, y_n)$$

The key of Euler's Method is to use the slope at  $f_n$  as the approximation slope of the entire step ( $\Delta x$ ), it's quite like a common sense lol. So Euler's Method underestimates when the actual function concaves up and overestimates when the function concaves down.

# Methodology and Experiment

The system of differential equations we use in this project can be written as:  
(*The derivation process is available in the appendix*).

$$\alpha_1 = \frac{d\omega_1}{dt} = \frac{-g(2m_1 + m_2) \sin(\theta_1) - m_2 g \sin(\theta_1 - 2\theta_2) - 2 \sin(\Delta) m_2 (\omega_2^2 l_2 + \omega_1^2 l_1 \cos(\Delta))}{l_1 \cdot [2m_1 + m_2 - m_2 \cos(2\theta_1 - 2\theta_2)]}$$
$$\alpha_2 = \frac{d\omega_2}{dt} = \frac{2 \sin(\Delta) [\omega_1^2 l_1 (m_1 + m_2) + g(m_1 + m_2) \cos \theta_1 + \omega_2^2 l_2 m_2 \cos \Delta]}{l_2 \cdot [2m_1 + m_2 - m_2 \cos(2\theta_1 - 2\theta_2)]}$$

Where,

$$\omega_1 = \frac{d\theta_1}{dt}, \omega_2 = \frac{d\theta_2}{dt}, \Delta = \theta_1 - \theta_2$$

In this system,  $\theta_\bullet$ ,  $\omega_\bullet$ , and  $\alpha_\bullet$  respectively represents angular position, angular speed and angular acceleration for each part of the pendulum.  $m_\bullet$  is the symbol of the mass of each pendulum, and  $l_\bullet$  is the length of each pendulum.

# Methodology and Experiment

- ▶ **Experiment 1:** Simulating double pendulum motions with slight initial difference, observing and discovering the *chaos* in such systems. Visualize experimental results, and describe.
- ▶ **Experiment 2:**